

Experiment : 1

Aim :

Introduction to Computer Hardware: Physical identification of major components of a computer system such as motherboard, RAM modules, daughter cards, bus slots, SMPS, internal storage devices, interfacing ports.

Specifications of desktop and server class computers. Installation of common operating systems for desktop and server use. (formulate specification for computer to be used as Desktop Web Server).

Prepare description and image for the following:

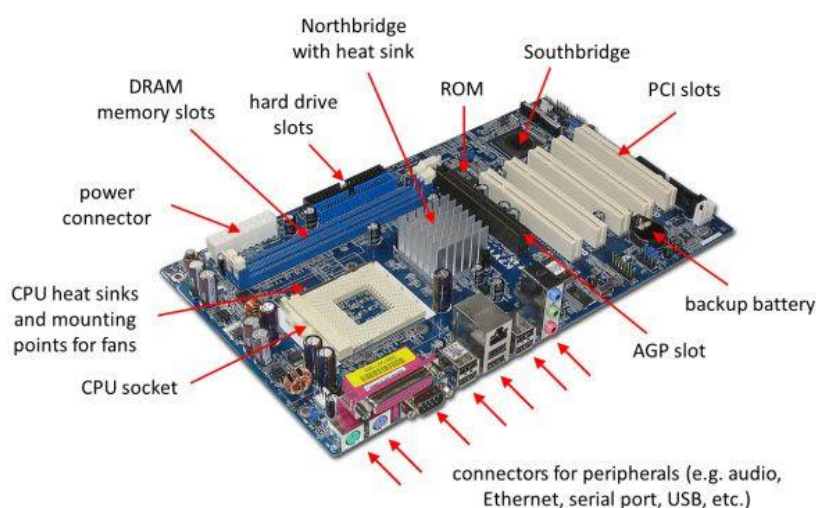
1. Motherboard

A motherboard is the control circuit board of a computer that connects and integrates all essential components, including the CPU, RAM, storage and peripherals. It provides communication pathways, power distribution, and expansion slots for additional hardware. The chipset on the motherboard controls data flow between components, ensuring efficient performance.

Key components of a motherboard:

CPU Socket, Chipset, RAM Slots, Expansion Slots (PCIe, PCI, etc.), Storage Connectors, Power Connectors, I/O Ports (USB, HDMI, Audio, Ethernet etc.), BIOS/UEFI firmware.

Motherboards come in various models designed for gaming, workstation, and general use, with different features like RGB lighting, overclocking support, Wi-Fi, etc.



2. RAM Modules

RAM (Random Access Memory) modules are high-speed memory components that temporarily store data and instructions for quick CPU access. They come in different types such as DDR4 and DDR5, with varying speeds and capacities, typically ranging from 4GB to 64GB per module in consumer systems. RAM modules plug into the motherboard's DIMM (Dual In-line Memory Module) slots in desktop computers and significantly impact system performance by improving multitasking and application speed. Faster and larger RAM can enhance system responsiveness, making it important for gaming, video editing, and other intensive tasks.



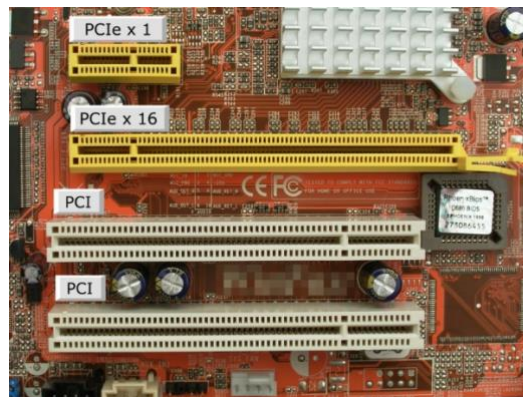
3. Daughter cards

A daughter card (or daughterboard) is a circuit board that connects to a motherboard or main board to provide additional functionality. It is typically used to expand features such as graphics processing, sound, network connectivity, or storage interfaces. Daughter cards attach via slots, connectors, or ribbon cables and are commonly found in laptops, servers, and embedded or modular hardware systems. Unlike standalone expansion cards, they often depend on a primary board for power and communication, making them a space-efficient way to enhance system capabilities.



4. Bus slots

Bus slots are expansion interfaces on a motherboard that allow additional hardware components, such as graphics cards, sound cards, network adapters, and storage controllers, to be installed. These slots provide data transfer pathways between the motherboard and the installed components or peripherals. Common types include PCI Express (PCIe), which comes in different lane configurations (x1, x4, x8, x16) offering varying bandwidth and performance levels, as well as older standards like PCI and AGP.



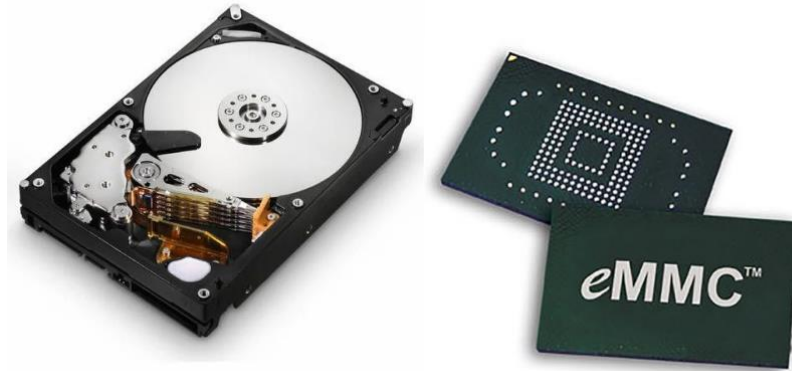
5. SMPS

SMPS (Switched-Mode Power Supply) is an electrical device that converts AC (alternating current) from the mains power supply into regulated DC (direct current) voltages required by a computer and other electronic devices. It is more efficient than traditional linear power supplies because it uses high-frequency switching to minimize energy loss. SMPS units provide multiple output voltages, such as +12V, +5V, and +3.3V, which are used by the motherboard and other components. These voltages are further distributed to parts like the CPU, GPU, and storage devices through the motherboard's power regulation system.



6. Internal Storage Devices

Internal storage devices are hardware components used to store the operating system, applications, and user data within a computer system. The two main types are HDDs (Hard Disk Drives), which use spinning magnetic platters to provide large-capacity, cost-effective storage, and SSDs (Solid State Drives), which use flash memory to deliver significantly faster performance, lower power consumption, and greater durability.



Hard disk drives

eMMC



NVMe SSD

7. Interfacing ports

Interface ports are physical connectors on a computer or electronic device that allow communication between the system and external devices. Common ports include USB (Universal Serial Bus) for connecting peripherals such as keyboards, mice, and storage devices, and for data transfer and power delivery; HDMI (High-Definition Multimedia Interface) for transmitting video and audio signals to displays; Ethernet ports for wired network connections; audio jacks for speakers and headphones; and DisplayPort for connecting high-resolution monitors.



Types of ports:

1. **Serial Ports:** Serial ports transmit data one bit at a time over a single communication channel. They are commonly used in older devices such as modems and legacy mice. They typically follow the RS-232 standard and are used for low-speed serial communication.
2. **Parallel Ports:** Parallel ports transmit multiple bits of data simultaneously over multiple channels. They were once widely used for connecting printers and other peripherals. They usually feature 25-pin or 36-pin connectors but are now largely obsolete, having been replaced by USB interfaces.

Desktop Computer Specifications

A typical desktop computer specification includes a multicore CPU (such as Intel Core i5/i7 or AMD Ryzen), 8GB to 32GB of RAM for smooth multitasking, and a 1TB HDD or 512GB+ SSD for storage, with SSDs providing faster boot and load times. Graphics processing is handled by either an integrated GPU or a dedicated graphics card from NVIDIA or AMD for gaming or professional applications. The system is built around a motherboard that provides various connectivity ports such as USB, HDMI, and Ethernet, and is powered by a reliable 500W to 750W SMPS. It also includes Wi-Fi and Bluetooth for wireless connectivity, along with support for peripherals like monitors, keyboards, and speakers.

Server Class Computer Specifications

A server-class computer typically features a multi-processor configuration with high-performance Intel Xeon or AMD EPYC CPUs, often offering 16 to 64 cores for handling demanding workloads. It is equipped with 32GB to 512GB of ECC (Error Correcting Code) RAM for enhanced stability and data integrity. Storage usually consists of RAID-configured HDDs or high-capacity SSD arrays, providing redundancy and fast data access. Servers use enterprise-grade motherboards that support multiple expansion cards, high-speed 10GbE or higher network interfaces, and redundant power supplies to ensure continuous uptime. These systems are designed for high reliability, scalability, and performance in demanding business and data center environments.

OS Installation Steps:

1. **Prepare Installation Media:** Create a bootable USB or DVD containing the operating system.
2. **Boot from Media:** Restart the computer and set the system to boot from the USB/DVD using BIOS/UEFI settings.
3. **Start Installation:** Follow the on-screen instructions to begin the installation process.
4. **Partition and Format Drive:** Select the storage drive, create partitions if needed, and format it for installation.
5. **Install OS:** Allow the operating system to install; the system may restart several times during the process.
6. **Complete Setup:** Configure user preferences such as language, region, account details, and install necessary updates and drivers.

Experiment : 2

Aim :

Familiarization of basic linux commands

Basic Linux commands, familiarity with following common ds/operations expected.

1. man

man is the system's manual page viewer. Each page argument given to man is normally the name of a program, utility, or function.

Options:

-d, --debug : Prints debugging information.

Example:

```
man ls
```

Displays the manual page for the program

2. ls, echo, read

- **ls:** Lists information about files (current directory by default).

Options:

-a, --all : Do not ignore entries starting with .

Example:

```
ls
```

- **echo:** Displays a line of text. Echoes the string(s) to standard output.

Options:

-n : Do not output the trailing newline.

Example:

```
echo "Hello, Linux!"
```

- **read:** Reads input from standard input (keyboard) or a file.

Options:

-p : Displays a prompt before reading input.

Example:

```
read name
```

```
echo "Hello, $name"
```

3. more, less, cat

- **more:** More is a filter for paging through text one screen at a time.

Options:

-s, --squeeze : Squeezes multiple blank lines into one.

Example:

```
more filename
```

- **less:** Less is a program similar to more, but with more features. It does not have to read the entire input file before starting, so it works faster with large files.

Options:

-c, --clear-screen : Repaints the screen from the top instead of scrolling from the bottom.

Example:

```
less filename
```

- **cat:** Displays the entire file content.

Options:

-A, --show-all : Equivalent to -vET.

-b, --number-nonblank : Numbers non-empty output lines.

-n, --number : Numbers all output lines.

Example:

```
cat filename
```


4. cd, mkdir, pwd, find

- **cd**: Changes the current working directory to dirName, or to the home directory if dirName is not given.

Example:

```
cd ~fred :Change to the home directory of the user fred
```

- **mkdir**: Creates directories if they do not already exist.

Options:

-p, --parents : No error if directory exists; creates parent directories as needed.

Example:

```
mkdir -p my_folder
```

- **pwd**: Prints the full path of the current working directory.

Options:

-L, --logical : Uses PWD from environment, even if it contains symbolic links.

Example:

```
pwd -L
```

- **find**: Searches for files and directories.

Options:

-depth : Processes directory contents before the directory itself (option -d is a synonym on some systems).

Example:

```
find / -name "file.txt"
```

5. mv, cp, rm, tar

- **mv**: Moves one or more files or directories from one location to another.

Options:

-n : Prevents overwriting an existing file.

Example:

```
mv old.txt new.txt
```

- **cp:** Copies files and directories.

Options:

-i : Prompts before overwriting an existing file.

Example:

```
cp -i source.txt destination.txt
```

- **rm:** Removes files or directories.

Options:

-i : Prompts for confirmation before removing each file.

Example:

```
rm -i file.txt
```

- **tar:** Creates and extracts archive files.

Options:

-c : Creates a new archive.

Example:

```
tar -cvf archive.tar folder/
```

6. wc, cut, paste

- **wc:** Used to count the number of lines, words, bytes, and characters in a file.

Options:

-w : Prints the number of words in a file.

Example:

```
wc file.txt
```

- **cut:** Used to extract sections from each line of a file and display them.

Options:

-b LIST : Selects only the bytes specified in LIST.

Example:

```
cut -b 1-5 file.txt
```

- **paste:** Joins files horizontally by merging corresponding lines, separated by a tab.

Options:

-s, --serial : Merges files sequentially.

Example:

```
paste file1.txt file2.txt
```

7. head, tail, grep, expr

- **head:** Prints the first n lines of a file (default is 10).

Options:

-n num : Prints the first num lines.

Example:

```
head file.txt
```

- **tail:** Prints the last n lines of a file (default is 10).

Options:

-n num : Prints the last num lines.

Example:

```
tail file.txt
```

- **grep:** Searches for a pattern in a file and displays matching lines.

Options:

-c : Prints only the count of matching lines.

Example:

```
grep "error" logfile.txt
```

- **expr**: Evaluates expressions and displays the result.

Options:

--version : Shows version information.

Example:

```
expr 12 + 8
```

8. chmod, chown

- **chmod**: Modifies file and directory permissions.

Options:

-f : Suppresses most error messages.

Examples:

```
chmod 755 script.sh
```

```
chmod u+rw filename
```

- **chown**: Changes file owner or group.

Options:

-c : Reports when a change is made.

Example:

```
chown -c mya masterfile1.txt
```

9. Redirections & Piping

- **>** Redirects output to a file (overwrites if it exists).

Example:

```
ls > output.txt
```

- **>>** Appends output to a file.

Example:

```
echo "Hello" >> file.txt
```

- **|** Pipes output from one command to another.

Example:

```
ls | grep ".txt"
```

10. useradd, usermod, userdel, passwd

- **useradd:** Adds a new user account.

Options:

-o, --non-unique : Allows duplicate UID (used with -u).

-u, --uid UID : Specifies user ID.

Example:

```
sudo useradd test_user
```

- **usermod:** Modifies an existing user account.

Options:

-aG : Adds user to a supplementary group.

Example:

```
usermod -aG groupname username
```

- **userdel:** Deletes a user account.

Options:

-r : Removes user along with home directory.

Example:

```
userdel -r username
```

- **passwd:** Sets or changes a user password.

Options:

-l : Locks the user account.

Example:

```
passwd username
```

11. df, top, ps

- **df:** Displays disk space usage of file systems.

Options:

-h : Shows output in human-readable format.

Example:

```
df -h
```

- **top:** Displays real-time system processes and resource usage.

Options:

-u : Shows processes of a specific user.

Example:

```
top -u username
```

- **ps:** Displays information about active processes.

Options:

-aux : Shows detailed information of all running processes.

Example:

```
ps aux
```

12. ssh, scp, ssh-keygen, ssh-copy-id

- **ssh:** Connects securely to a remote machine.

Options:

-p : Specifies the port number.

Example:

```
ssh -p 22 user@host
```

- **scp:** Copies files securely between systems.

Options:

-r : Copies directories recursively.

Example:

```
scp -r file.txt user@host:/path/
```

- **ssh-keygen:** Generates SSH key pairs.

Options:

-t : Specifies the key type.

Example:

```
ssh-keygen -t rsa
```

- **ssh-copy-id:** Copies SSH public key to a remote server.

Options:

-i : Specifies the identity file.

Example:

```
ssh-copy-id -i ~/.ssh/id_rsa.pub user@host
```

Experiment : 3

Aim :

File system hierarchy in a common Linux distribution, file and device permissions, study of system configuration files, familiarizing log files for system events, user activity, and network events.

1. File System Hierarchy in a Common Linux Distribution

The Linux file system follows a standardized directory structure called the File System Hierarchy Standard (FHS). This structure defines the organization of system files, applications, and directories. The hierarchy is arranged as a tree, starting with the root directory (/), and branching into various subdirectories that contain specific types of files.

Here is a detailed look at the most common directories and their purposes:

- **/ (Root Directory):**

The top-level directory of the entire file system. All other directories and files are contained within this directory.

- **/bin:**

Contains essential user commands (like ls, cp, mv, etc.) required for basic system operations

- **/boot:**

Contains files required for system booting, including the kernel (vmlinuz), bootloader configuration files, and the initial RAM disk (initrd).

- **/dev:**

Contains device files that represent hardware and virtual devices used by the operating system.

- **/etc:**

Stores system-wide configuration files used by the system and administrators.

- **/home:**

Contains users' personal directories. Each user has a separate subdirectory named after their username.

- **/lib:**

Contains essential shared libraries required by system programs to run.

- **/media:**

When removable media devices (like USB drives or CDs) are inserted, they are typically mounted automatically under the /media directory in subdirectories (e.g., /media/usb, /media/cdrom). It provides access to removable storage devices.

- **/mnt:**

/mnt is used for temporary mounting of external or network file systems, usually for administrative purposes.

- **/opt:**

Contains optional or third-party software packages that are not part of the default system installation.

- **/proc:**

A virtual filesystem that provides real-time information about the system, processes, and kernel.

- **/root:**

The home directory of the root (superuser) account.

- **/run:**

A temporary filesystem that stores runtime data needed during system operation. This data is cleared on reboot.

PID files: Store process IDs of running services.

Lock files: Prevent multiple instances of the same process from running simultaneously.

- **/sbin:**

Contains essential system administration commands, mainly used by the root user.

- **/srv:**

Stores data for services provided by the system (e.g., web or FTP servers).

- **/sys:**

A virtual file system that provides information about hardware and allows interaction with kernel parameters.

- **/tmp:**

Stores temporary files created by applications. These files are usually deleted after reboot.

- **/usr:**

Contains user programs, libraries, documentation, and non-volatile data required for normal system operation.

- **/var:**

Contains variable data that changes frequently, such as logs, databases, emails, and cache files.

Example: logs : Stores system and application log files.

2. File and Device Permissions

Linux uses a permission model to control access to files and devices. Permissions are granted at three levels: user, group, and others.

Basic Permissions:

- **Read (r):** Permission to read the contents of a file or list the contents of a directory.
- **Write (w):** Permission to modify or delete a file, or add/remove files in a directory.
- **Execute (x):** Permission to run a file as a program or script, or to access (search) a directory.

Permission Representation: Permissions are displayed using a 10-character string:

`-rwx rwx r--`

- The first character indicates the file type (e.g., - for regular file, d for directory).
- The next nine characters represent permissions.
- These are divided into three groups of three:
 - o User (owner) permissions
 - o Group permissions
 - o Others permissions

Changing Permissions:

- **chmod:** Used to change file or directory permissions.

Changing Ownership:

- **chown:** Used to change the owner of a file or directory.

3) System Configuration Files in /etc

The /etc directory contains configuration files for the system and its applications. Key files include:

- **/etc/passwd:** Contains basic account information
- **/etc/group:** Defines group information
- **/etc/fstab:** Defines how disk partitions and devices are mounted
- **/etc/network/interfaces:** Network interface configuration (used in older systems)
- **/etc/hostname:** Contains the system's hostname
- **/etc/resolv.conf:** Configures DNS servers for the system
- **/etc/crontab:** Contains cron jobs for scheduled tasks .It is used to allow
 - administrative use of cron and needs to be run periodically (e.g., hourly, daily)
- **/etc/services:** Contains service-to-port mappings

4) System Log Files for Events, User Activity, and Network Events

Linux maintains various log files in /var/log that capture system activities, user actions, and network events.

Important Log Files:

- **/var/log/syslog:** Contains general system activity logs.
- **/var/log/auth.log:** Logs authentication-related events.
- **/var/log/messages:** Contains general system messages.
- **/var/log/kern.log:** Contains kernel-related messages.
- **/var/log/secure:** Logs security-related events.
- **/var/log/boot.log:** Contains messages logged during system boot.
- **/var/log/cron:** Logs scheduled tasks executed via cron.
- **/var/log/dmesg:** Contains kernel ring buffer messages.

- **/var/log/lastlog:** Stores last login times of users.

Monitoring Network Events:

- **/var/log/messages:** May include network-related logs.
- **/var/log/daemon.log:** Contains logs for system daemons handling network services.

A daemon is a background process that runs continuously and performs system-related tasks without direct user interaction. It is essential for managing services like logging, networking, and scheduling.

Tree Command:

The tree command displays the contents of a directory recursively, showing subdirectories and files along with a summary of the total number of sub-directories and files.

Enable the printing of hidden files using `-a` flag

Examples:

`sudo tree -a` (Displays hidden files as well)

`sudo tree -f -p` (Shows full path along with file type and permissions)

To install tree:

`sudo apt install tree`

